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PROJECT PROPOSAL
**TITLE: VIRTUALIZED AR WINDSCREEN DISPLAY WITH SEMI-
AUTONOMOUS DRIVING SUPPORT SYSTEM**

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Abbreviations and Acronyms

Abbreviation	Meaning
ADAS	Advanced Driver-Assistance System
AEB	Autonomous Emergency Braking
AR	Augmented Reality
CAN	Controller Area Network (in-vehicle bus)
CPA	Closest Point of Approach
CRC	Cyclic Redundancy Check
CTRV	Constant Turn Rate and Velocity (motion model)
CUDA	Compute Unified Device Architecture (NVIDIA GPU computing platform)
CV	Constant Velocity (motion model)
EKF	Extended Kalman Filter
FCW	Forward Collision Warning
FreeRTOS	Free Real-Time Operating System
GNSS / GPS	Global Navigation Satellite System / Global Positioning System
GPU	Graphics Processing Unit
HMM	Hidden Markov Model
HUD	Head-Up Display
I2C	Inter-Integrated Circuit (serial bus)
IMM	Interacting Multiple Model (filter)
IMU	Inertial Measurement Unit
IWDG	Independent Watchdog (STM32 hardware timer)
LiDAR	Light Detection and Ranging
mAP	mean Average Precision
MLD	Multi-Level Dijkstra (OSRM routing algorithm)
MTSD	Mapillary Traffic Sign Dataset
NTSA	National Transport and Safety Authority (Kenya)
OBD-II	On-Board Diagnostics, second generation
OSM	OpenStreetMap
OSRM	Open Source Routing Machine
PCB	Printed Circuit Board
PWM	Pulse-Width Modulation
TensorRT	NVIDIA's deep-learning inference optimiser and runtime
TTC	Time to Collision
UART	Universal Asynchronous Receiver-Transmitter
USB-CDC	USB Communications Device Class (virtual serial port)
WSL2	Windows Subsystem for Linux, version 2
YOLO	You Only Look Once (single-stage object detector family)

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Abstract

Road crashes killed an estimated 1.19 million people worldwide in 2021 [1], and Kenya records over four thousand road deaths a year, most of them pedestrians and motorcyclists [2]. Many of these crashes begin with a driver who looked away at the wrong moment: at a phone used for navigation, or at a hazard noticed too late. Driver-assistance technology that addresses this, such as augmented-reality (AR) head-up displays and automatic emergency braking, is sold almost exclusively as factory equipment in new, expensive vehicles, and none of it can be fitted to the older cars that make up most of Kenya's fleet.

This project proposes a **Virtualized AR Windscreen Display with Semi-Autonomous Driving Support System**: a retrofit platform that fuses a 2K camera, a 360° LiDAR, satellite positioning, an inertial sensor and the car's own OBD-II data into a real-time model of the road ahead. The model drives two things. First, a **precision AR navigation display**, in which the route is drawn onto the live camera view locked to the measured road surface and to the correct lane, rather than as a generic arrow floating over the image. Second, a **semi-autonomous safety layer** that detects and tracks road users, warns of hazards and reckless driving, operates the indicators and hazard lights, and, as a last resort, applies a force-limited brake through a fail-safe actuator which the driver can always override.

The system is split between a laptop with an NVIDIA RTX 5060 GPU, which runs four TensorRT neural networks (object segmentation, Kenyan road-sign detection, lane detection and depth), sensor fusion, tracking, offline routing for a 100 km region around Nairobi and GPU rendering; and an STM32 microcontroller hub running FreeRTOS, built on two custom circuit boards: a windscreen pod behind the mirror carrying the camera and inertial sensor, and an under-dash power box whose outputs all default to "off" in hardware. The hub owns every actuator, reads the car's own CAN bus, and enforces the safety rules independently of the laptop, including release within 200 ms if the laptop stops responding. Much of the software has already been built and tested against recorded road video and a recorded Nairobi drive: the four networks run together in a median 17.8 ms per frame against a 33 ms budget, the AR renderer runs at 54–82 frames per second at 2560×1440, and the braking logic fires at the intended time-to-collision in simulation. The remaining work is hardware integration, calibration, a mandatory actuator bench test, and staged validation from a bench, to a stationary car, to an empty field, to a closed campus road at 5–10 km/h. The estimated hardware budget is KES 195,000–270,000, most of which is the LiDAR.

Keywords: augmented reality, head-up display, driver assistance, sensor fusion, LiDAR, deep learning, map matching, emergency braking, functional safety, embedded systems, CAN bus.

1.0 Introduction and Background

Driving is mostly a visual task. A driver continuously reads the road ahead, the vehicles and people around them, the signs, and, increasingly, a navigation screen mounted on the dashboard or held in the hand. Every glance away from the road is a period in which the driver is effectively blind to what happens in front of the car, and naturalistic driving studies have repeatedly found that such glances are among the strongest predictors of a crash [3][4].

Two families of technology have grown up to address this. The first is the **head-up display (HUD)**, which places information in the driver's line of sight so that the eyes do not leave the road. Its most advanced form, the **augmented-reality (AR) HUD**, goes further: it draws graphics that appear attached to the real world, such as a navigation path lying on the correct lane or a highlight around a pedestrian, so that the driver does not need to translate a map or an abstract symbol into the scene in front of them [5][6]. The second family is **advanced driver-assistance systems (ADAS)**: forward collision warning, lane keeping, and autonomous emergency braking (AEB), which use cameras and radar to detect danger and, in the case of AEB, brake the car when the driver does not. Large studies of AEB have found it reduces rear-end crashes substantially [7].

Both families share a limitation that matters in Kenya: they are factory-fitted to new, mid-range and premium vehicles and cannot be added afterwards. Commercial AR windscreen products, such as those demonstrated by WayRay and by vehicle manufacturers, are proprietary, costly and tied to specific car models. Most cars on Kenyan roads are older imports that will never receive them. At the same time, the components from which such a system is built (cameras, compact LiDAR sensors, satellite receivers, inertial sensors, consumer GPUs that can run neural networks in real time, and inexpensive microcontrollers) have become affordable and widely available.

This project proposes to combine those components into a **retrofit** platform: a **Virtualized AR Windscreen Display with Semi-Autonomous Driving Support System**. It is “virtualized” because, instead of an optical combiner projected on the glass, the first version shows the AR view on a screen as a live camera image with the graphics composited into it (video see-through AR), while the software is structured so that an optical display can replace the screen later without changing anything else. It is “semi-autonomous” because the car never drives itself: the system informs the driver, operates the indicators, hazard lights and horn, and, only when a collision is imminent and the driver has not reacted, applies a limited brake that the driver can always overpower.

The central idea is that a single, precise, real-time model of the road ahead serves both purposes. The same sensor fusion that tells the navigation display exactly where the road surface and the lane are also tells the safety layer where every vehicle, cyclist and pedestrian is, how fast they are moving and where they are likely to go next.

1.1 Problem Statement

Kenya's National Transport and Safety Authority recorded 4,690 road deaths in 2022, of whom 1,682 were pedestrians and 1,254 were motorcyclists [2]. Globally the World Health Organization estimates 1.19 million deaths in 2021, with more than nine in ten occurring in low- and middle-income countries [1]. Three driver-side problems contribute to this toll and are the focus of this project:

1. **Delayed recognition of hazards.** Pedestrians crossing between vehicles, motorcycles filtering through traffic, vehicles braking suddenly and vehicles driven recklessly are recognised late, especially at night, in rain and in dense urban traffic. The time the driver has left to react is often shorter than their reaction time.
2. **Distraction caused by navigation.** Drivers who do not know the route glance repeatedly at a phone or a dashboard screen and must mentally map a two-dimensional arrow onto a complex junction. Each glance removes their eyes from the road; the 100-car naturalistic study found that glances away from the forward roadway of more than two seconds roughly doubled near-crash and crash risk [3]. Kenyan roads add a further difficulty: road works, closures and new roads change the network faster than a static map can follow.
3. **No automatic protective response.** In the vehicles most Kenyans drive, nothing reacts on the driver's behalf when a collision becomes imminent: no warning, no hazard lights to alert following traffic, no emergency braking.

Solutions to each problem exist, but only as factory-fitted, model-specific and expensive equipment. There is therefore no affordable, retrofittable system that gives the driver of an ordinary car precise, eyes-on-the-road navigation and hazard awareness, together with a safe, bounded automatic response when the driver does not react in time.

1.2 Problem Justification

Solving this problem is worthwhile for four reasons.

The need is large and local. The groups most at risk on Kenyan roads, pedestrians and motorcyclists, are exactly those a camera–LiDAR perception system can detect and highlight. Even a modest reduction in late recognition of such road users would have real value.

The evidence supports the approach. AR navigation that places guidance on the road reduces the need to look away, and drivers using it have reported clear benefits over a conventional display [6], provided the graphics are designed not to become a distraction themselves [8]. Forward collision warning combined with automatic braking has been associated with a large reduction in front-to-rear crashes with injuries [7].

A retrofit is now technically and economically feasible. A consumer laptop GPU can run several neural networks on high-resolution video in real time, a compact LiDAR costs a fraction of earlier automotive units, and open data and software (OpenStreetMap, OSRM, TensorRT, FreeRTOS) remove licensing costs. The whole system is estimated at KES 195,000–270,000, far below the cost of a vehicle equipped with comparable factory features.

It is a substantial engineering project in the department's core areas. It spans embedded systems and real-time firmware, power and actuator electronics, communication protocols, digital signal processing and estimation, computer vision and machine learning, computer graphics, and safety engineering, which is the breadth expected of a final year project in Electronic and Computer Engineering.

1.3 Objectives

1.3.1 Main Objective

To design, implement and test a precision, sensor-fused augmented-reality navigation and driver support system, whose real-time model of the road also drives hazard detection and fail-safe, driver-overrideable vehicle actuation, as a retrofit for an ordinary car.

1.3.2 Specific Objectives

1. To design and implement a real-time perception and sensor-fusion pipeline that detects, segments and ranges vehicles, pedestrians, cyclists, lanes and Kenyan road signs from camera and LiDAR data, and estimates the car's own motion from GNSS, IMU and OBD-II, processing 2K video at 30 frames per second (all neural networks within a 33 ms frame budget) with a road-sign detection accuracy of at least 0.70 mAP50.
2. To design and implement a road-locked AR navigation display that routes offline across a 100 km region around Nairobi, matches the car's position to the road, and draws the route in the correct lane on the measured road surface at 24 frames per second or more, with the drawn path within 2 pixels of the calibrated camera model and visibly inside the correct lane within LiDAR range.
3. To design, implement and validate a hazard-assessment and fail-safe actuation subsystem that warns of hazards and reckless driving, operates indicators, hazard lights and horn, and brakes through a force-limited actuator only when time-to-collision falls below 1.8 s and the driver has not braked, verified by passing all six actuator bench checks on three consecutive attempts (including driver override, release within 200 ms of link loss, and release when the power box is disconnected) before staged in-vehicle testing at 5–10 km/h.

1.4 Scope and Limitations

In scope. The full sensing, perception, fusion, tracking, navigation, decision and rendering software on the laptop; the sensor and actuator hub (firmware, and two custom PCBs with their enclosures: a windscreen pod and an under-dash power box); a roof mount for the LiDAR; a force-limited brake actuator with a fail-safe release; a physical emergency-stop switch; a bench rig for the actuator; installation in one test vehicle; and staged validation up to a closed campus road section at 5–10 km/h.

Out of scope for this project, and presented as future work: an optical combiner or projector on the windscreen (the software's display interface is designed so this can be added without other changes); steering intervention of any kind; fully autonomous driving; thermal cameras; and cloud connectivity or mobile applications.

Limitations. Navigation precision is highest within LiDAR range, where the road surface is measured, and falls back to a flat-road assumption beyond it; this will be reported as it is. Testing is limited to low speeds on closed or empty ground, so stopping-distance calibration at road speeds is not claimed. A live road-status layer (traffic and incidents) depends on a commercial data provider's licence terms and is deferred; the system routes on an offline OpenStreetMap graph that is refreshed regularly and corrected by the car's own sensors.

2.0 Literature Review

This chapter reviews the work on which each part of the proposed system builds: why driver attention and navigation matter (2.1); augmented-reality head-up displays (2.2); deep-learning perception from a camera (2.3); fusing the camera with LiDAR and estimating motion (2.4); routing and map matching (2.5); and collision avoidance and functional safety (2.6). Each section ends by stating what this project takes from the literature and where it differs. Section 2.7 draws the gaps together.

2.1 Road Safety, Driver Attention and Navigation Displays

The WHO's 2023 global status report estimates 1.19 million road deaths in 2021, a slight fall from 2010 but still the leading cause of death for people aged 5–29, with more than 90% of deaths in low- and middle-income countries and more than half among pedestrians, cyclists and motorcyclists [1]. Kenyan figures follow the same pattern: pedestrians and motorcyclists together made up more than 60% of the 4,690 deaths recorded in 2022 [2]. Any system meant for Kenyan roads must therefore detect vulnerable road users reliably, not only other cars.

The link between inattention and crashes is well established by *naturalistic driving studies*, in which ordinary drivers' cars are instrumented with cameras and sensors for months. Klauer *et al.* analysed the 100-car study and found that any glance away from the forward roadway lasting more than two seconds at least doubled the risk of a crash or near-crash, and that secondary tasks were a factor in a large share of events [3]. Dingus *et al.*, using the much larger SHRP 2 dataset of over 900 crashes, found that drivers were engaged in some observable distracting activity during more than half of their driving time, and that handling a hand-held phone raised crash risk several-fold [4].

These studies show why a conventional navigation display is itself part of the problem: to use it, the driver must look down, read an abstract map, and translate it back to the scene. Placing guidance in the driver's line of sight and attaching it to the road is the obvious response, and leads to head-up and AR displays.

2.2 Augmented-Reality Head-Up Displays

Gabbard, Fitch and Kim reviewed the prospects for automotive AR and set out both its promise and its difficulties [5]. The promise is that graphics *registered* to the world (appearing to lie at the correct place in the scene) remove the mental translation step: a turn arrow painted on the correct lane needs no interpretation. The difficulties are technical and human. Registration must be accurate and stable, or the graphics drift, jitter or appear to float, which is worse than no graphics at all. The graphics must not hide real hazards. And the system's latency must be low enough that the overlay keeps up with the car's motion.

Bark *et al.* built "Personal Navi", a navigation aid on a see-through 3D volumetric HUD, and compared AR guidance that appeared at the depth of the road with conventional guidance in a driving study [6]. The study reported advantages for guidance presented at the depth of the road, supporting the idea that guidance attached to the road reduces the effort of following a route. Their system, however, relied on a specialised optical display and on a pre-defined route geometry, not on sensing the road itself.

The risk of AR adding distraction rather than removing it is addressed by Kim and Gabbard, who measured how different AR HUD designs affect drivers’ visual attention [8]. They found that the distraction potential depends strongly on how the graphics are designed and placed, and that it can and should be measured rather than assumed. This project turns that finding into design rules: one consistent visual language, graphics that stay on the road surface, graphics shown only when relevant, and no overlay covering a hazard.

Commercial systems follow the optical route. Production AR HUDs from vehicle manufacturers, and holographic windscreen displays shown by companies such as WayRay, project graphics through the windscreen with an optical combiner. They are engineered for a specific car’s windscreen and dashboard, are proprietary, and are not available as retrofits.

Relevance to this project. The proposal adopts the AR principle and the design rules from [5][8], but makes two different choices. First, it uses *video see-through* AR on a screen for the first version, which removes the optical engineering while keeping the registration problem, the hard part. Second, the registration is driven by *measurement*: the route is placed using the LiDAR-measured road surface and the detected lane lines, rather than a flat-road assumption or pre-defined geometry.

2.3 Camera-Based Perception with Deep Learning

Object detection and segmentation. Redmon *et al.* introduced YOLO, which treats detection as a single regression from the image to boxes and class probabilities and so runs in real time, unlike earlier two-stage detectors [9]. The family has since gone through many versions; Terven, Córdova-Esparza and Romero-González review them up to YOLOv8, whose anchor-free head and improved training give a good balance of speed and accuracy [10]. Bolya *et al.* showed with YOLACT that *instance segmentation*, a pixel-accurate mask for each object, can also run in real time, by predicting a small set of prototype masks per image and a set of coefficients per object [11]. YOLOv8’s segmentation variant uses the same idea. Masks matter for this project for two reasons: the AR highlight can follow a pedestrian’s outline, and, more importantly, the mask selects exactly which LiDAR points belong to the object (Section 2.4).

Lane detection. Qin, Wang and Li reformulated lane detection as selecting, for each of a set of predefined image rows, which column the lane passes through, rather than classifying every pixel [12]. This “row anchor” approach is far faster than segmentation and copes with lanes hidden by vehicles because it reasons about the whole image. Their follow-up, UFLDv2, adds column anchors and an ordinal classification loss, improving accuracy on curved and side lanes while keeping the speed [13].

Monocular depth. Ranftl *et al.* trained MiDaS on a mixture of datasets with a scale- and shift-invariant loss, producing a network that predicts *relative* depth for almost any image [14]. Its output has no metric scale, so it cannot on its own give a distance in metres, but it is useful for ordering objects and, when scaled against a few measured points, for estimating the range of objects the LiDAR does not see.

Traffic signs. Ertler *et al.* released the Mapillary Traffic Sign Dataset (MTSD), with over 100,000 street-level images from around the world and more than 300 sign classes [15]. Its global coverage makes it a better starting point for Kenyan roads than datasets collected in one European country, since Kenyan signs follow the Vienna-convention style with local variations.

Relevance to this project. Detection alone gives *what* and *where in the image*, not *how far*. Most published systems evaluate one network at a time on a desktop GPU. This project runs four networks together (YOLOv8m-seg, a YOLOv8 sign detector trained on MTSD and audited on Kenyan images, UFLDV2 and MiDaS) within one frame period on a laptop GPU, and uses each network’s output as an input to geometry and fusion rather than as an end result.

2.4 Camera–LiDAR Fusion, State Estimation and Tracking

Geometry and calibration. Fusing a camera with any 3D sensor starts from the pinhole camera model and its lens distortion, which map a 3D point to a pixel [16]. Zhang’s method estimates these parameters from several views of a flat checkerboard and is the standard procedure for camera calibration [17]. The rigid transform between LiDAR and camera (the *extrinsics*) must also be known; small errors in it move projected points off the objects they belong to.

Fusion. Qi *et al.*’s Frustum PointNets use a 2D detection to cut out the frustum of 3D points behind each box and then find the object inside it with a learned network [18]. Vora *et al.*’s PointPainting goes the other way: it projects every LiDAR point into a semantic segmentation of the image and “paints” it with the class scores before 3D detection [19]. Both show that the camera is best at deciding *what* something is and the LiDAR at measuring *where* it is. Both, however, rely on trained 3D networks and on large labelled LiDAR datasets. Box-based selection also has a known weakness: a pedestrian’s bounding box is mostly background, so points from the wall behind them contaminate the range.

State estimation. The Kalman filter [20] is the optimal estimator for linear systems with Gaussian noise; the extended Kalman filter (EKF) linearises a nonlinear model around the current estimate, and the unscented filter of Julier and Uhlmann propagates sample points instead, which handles stronger nonlinearity [21]. Thrun, Burgard and Fox give the probabilistic framework, including motion models for vehicles and the fusion of satellite, inertial and odometry measurements [22].

Tracking manoeuvring targets. A single motion model cannot describe a car that cruises, turns and brakes. Blom and Bar-Shalom’s *interacting multiple model* (IMM) filter runs several models in parallel (for example constant velocity, constant turn rate, and stopping) and mixes them according to how well each explains recent measurements [23]. Tracking many objects adds the *data association* problem, deciding which detection belongs to which track, which is commonly solved as an assignment problem with the Hungarian algorithm [24] inside validation gates [25].

Relevance to this project. The proposal fuses with *instance masks* rather than boxes: only points inside an eroded mask are assigned to an object, and the range is taken from the nearest dominant cluster of those points, never their mean. This needs no trained 3D network and removes background contamination at its source. Tracking uses an IMM filter with constant velocity, constant turn rate and velocity (CTRV) and stopping models, which also provides a per-object probability that a vehicle ahead is braking.

2.5 Routing and Map Matching

Luxen and Vetter presented OSRM, an open-source routing engine for OpenStreetMap data that answers continental-scale route queries in milliseconds by preprocessing the road graph [26]. Delling *et al.*’s *customizable route planning* separates that preprocessing into a metric-independent partition of

the graph and a fast customisation step that applies road weights [27]. OSRM’s multi-level Dijkstra (MLD) mode implements this idea, which means that closures or traffic can re-weight the graph in seconds without rebuilding it.

Raw GNSS positions are typically in error by several metres and cannot say which road, let alone which lane, the car is on. Newson and Krumm formulated *map matching* as a hidden Markov model: candidate road positions near each fix are hidden states, emission probabilities fall with distance from the fix, and transition probabilities favour routes whose driven length matches the straight-line distance between fixes; the Viterbi algorithm then finds the most likely path [28]. This is robust to noise and to sparse fixes and is the basis of OSRM’s own matching service.

Relevance to this project. Routing and map matching establish *which road* and *how far along it* the car is, but not where the road is in the camera image with lane-level precision. The proposal therefore adds a step not found in these works: a projector that places the matched route on the LiDAR-measured road surface and corrects its heading and lateral position from the detected lane lines.

2.6 Collision Avoidance and Functional Safety

Lee showed that drivers regulate braking using *time to collision* (TTC), the current gap divided by the closing speed, which can be perceived directly from how fast an object’s image grows [29]. TTC has since become the standard trigger quantity in forward collision warning and AEB. Cicchino’s analysis of police-reported crashes in the United States found that vehicles with forward collision warning plus low-speed AEB had about half the rate of front-to-rear crashes of the same models without them [7].

A system that can apply the brakes is safety-critical. ISO 26262 sets out the functional-safety lifecycle for road-vehicle electronics: hazard analysis, safety goals, a safe state for every fault, independence between the function and its monitor, and verification evidence at each stage [30]. Two concerns stand out for a retrofit. The first is that a false activation, *phantom braking*, is itself a hazard. The second is that software on a general-purpose computer cannot be trusted to guarantee a safe state.

Relevance to this project. The proposal does not claim ISO 26262 compliance, which is beyond a student project, but adopts its principles: a defined safe state (brake released), an independent monitor (the hub, which enforces limits and releases on its own watchdog), a hardware kill path (the emergency stop), and an evidence log of every decision. To limit phantom braking, braking is triggered only by the measured TTC of a confirmed, LiDAR-ranged object, never by a prediction or a learned model, as recommended by the TTC literature [29].

2.7 Summary of Gaps and the Proposed Improvements

Table 1 compares the reviewed work with the proposed system. The works reviewed each solve one part of the problem well; none combines precise, measured AR registration with perception, tracking and a fail-safe actuation path in a system that can be retrofitted to an ordinary car.

Table 1: Comparison of existing approaches with the proposed system

Approach	What it does well	Gap for this problem	How this project differs
Commercial AR HUDs (factory, optical)	High-quality optical AR in the driver’s line of sight	Proprietary, expensive, model-specific; not retrofittable	Retrofit hardware; video see-through first, optical display interface designed in

AR navigation studies [6][8]	Show the benefit and the design rules of conformal graphics	Route geometry assumed or pre-defined; no sensing of the road	Route placed on the LiDAR-measured surface, corrected by detected lanes
Real-time detectors [9][11][13]	Fast, accurate image-space detection	No metric range; evaluated in isolation	Four networks together within one frame; outputs feed geometry and fusion
Learned camera–LiDAR fusion [18] [19]	Accurate 3D detection	Needs trained 3D networks and labelled LiDAR data; box-based selection mixes in background	Mask-based point assignment with nearest-cluster range; no 3D training needed
Routing and HMM map matching [26] [28]	Which road, how far along it	Metre-level error; no lane or surface information	Adds the road-surface projector; three road-status layers where sensors override the map
Factory FCW/AEB [7]	Proven crash reduction	Factory-only; closed design	Retrofit, force-limited, driver-overridable, with an open, auditable trigger rule and an evidence log

In summary, the improvements this project makes over existing work are: (i) **measured registration** of the AR route, from the combination of LiDAR ground model, lane detection and map matching; (ii) **mask-based fusion** that gives a reliable range to each detected object without training a 3D network; (iii) **one road model serving both navigation and safety**; and (iv) a **retrofit, fail-safe actuation path** in which an independent microcontroller, a hardware kill switch and a mechanical force limit each keep the brake safe even if the laptop fails.

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